GEOMETRY QUEST · WORLD 14

Congruence

WORKSHEET

Name _____ Date _____ Stars I give myself: ☆ ☆ ☆

1 Two triangles share these facts. Which rule proves they are congruent: SSS, SAS, ASA, AAS, RHS or “not enough”?

all three sides match _____

two sides and the angle between them match _____

two angles and the side between them match _____

a right angle, the hypotenuse and one other side match _____

all three angles match _____

two sides and an angle NOT between them match _____

2 △ABC ≅ △PQR. AB = 5 cm, BC = 7 cm and angle A = 40°. Find PQ, QR and angle P.

PQ = _____ **QR =** _____ **∠P =** _____

3 Draw two triangles that both have angles of 50°, 60° and 70° but are different sizes. What does this show?

4 Why does a garden gate have a diagonal piece of wood across it?

Adapted from Geometry for Elementary School, Wikibooks, CC BY-SA 3.0